

Module Title: Stochastic Calculus for Finance

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School of Mathematics

Semester Two 202425

Use of Gen AI (Generative Artificial Intelligence) instructions:

- This assessment is red category. AI tools cannot be used.

Calculator instructions:

- You are allowed to use a non-programmable calculator in this examination.

Dictionary instructions:

- A basic English dictionary is available to use. Raise your hand and ask an invigilator if you need it.

Exam information:

- There are 3 pages to this examination.
- There will be **2 hours 30 minutes** to complete this examination.
- This examination is worth 80% of the module mark.
- There are three questions in this exam paper. You must answer all three questions.
- Questions 1 and 3 count for 30 marks. Question 2 counts for 40 marks.
- The numbers in brackets indicate the marks available for each sub-question.
- You must show all your calculations.
- You must write your answers in the answer booklet provided. If you require an additional answer booklet, raise your hand so an invigilator can provide one.
- You must clearly state your name and Student ID Number in the relevant boxes on your answer booklet. Other boxes may be left blank.

1. (a) [8 marks] Let Ω be a set. Give the definition of a σ -algebra $\mathcal{F}$ on Ω .
- (b) [10 marks] Let $\Omega = \{1, 2, 3, 4, 5\}$. Consider the subset $\{1, 2\} \subset \Omega$. Denote by $\mathcal{F}$ the σ -algebra generated by $\{1, 2\}$, that is $\mathcal{F} = \sigma(\{1, 2\})$.
- List all the elements of $\mathcal{F}$.
 - Let $\mathcal{B}(\mathbb{R})$ denote the Borel σ -algebra on $\mathbb{R}$. Consider the function $X : \Omega \rightarrow \mathbb{R}$ given by

$$X(\omega) = \begin{cases} 5, & \text{if } \omega \in \{1, 2, 4\} \\ 2, & \text{otherwise.} \end{cases}$$

Is X $\mathcal{F}/\mathcal{B}(\mathbb{R})$ -measurable? Provide arguments justifying your answer.

- (c) [12 marks] Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \rightarrow \mathbb{R}$ and $X_n : \Omega \rightarrow \mathbb{R}$, $n \in \mathbb{N}$, be random variables.
- When do we say that X_n converges to X in probability, as $n \rightarrow \infty$?
 - When do we say that X_n converges to X in $L_1(\Omega)$, as $n \rightarrow \infty$?
 - Show that if X_n converges to X in $L_1(\Omega)$, as $n \rightarrow \infty$, then X_n converges to X in probability, as $n \rightarrow \infty$.
 - Suppose that there exist $A_n \in \mathcal{F}$ with $\mathbb{P}(A_n) = \frac{1}{n}$ such that

$$X_n(\omega) = \begin{cases} n^2, & \text{if } \omega \in A_n \\ \frac{1}{n}, & \text{otherwise.} \end{cases}$$

Show that X_n converges to 0 in probability, as $n \rightarrow \infty$. Does it also converge in $L_1(\Omega)$? Explain your answer.

2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $T > 0$, and consider a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$ consisting of σ -algebras $\mathcal{F}_t \subset \mathcal{F}$.
- (a) [8 marks] Give the definition of a one dimensional Wiener process $(W(t))_{t \in [0, T]}$ on $(\Omega, \mathcal{F}, \mathbb{P})$. When do we say that a Wiener process $(W(t))_{t \in [0, T]}$ on $(\Omega, \mathcal{F}, \mathbb{P})$ is an $\mathbb{F}$ -Wiener process on $(\Omega, \mathcal{F}, \mathbb{P})$?
- (b) [10 marks] Let $(W(t))_{t \in [0, T]}$ be a Wiener process and let $T = 100$.
- Compute the following quantities:

$$\mathbb{E}[(W(2))^2], \quad \mathbb{E}[W(7)W(2)], \quad \mathbb{E}[(W(7))^2 W(2)].$$

- Consider the function $g : [0, \infty) \rightarrow [0, \infty)$ given by $g(s) := \sqrt{s}$ for $s \in [0, \infty)$. Find the expectation and the variance of the random variable

$$\int_0^1 g(s) dW(s).$$

- (c) [14 marks] Let $(W(t))_{t \in [0, T]}$ be an $\mathbb{F}$ -Wiener process. For $t \in [0, T]$, let us set

$$Y(t) = \sin\left(\frac{\pi}{2} + W(t)\right).$$

- i. Find the stochastic differential of $(Y(t))_{t \in [0, T]}$.
- ii. Consider the process $(M(t))_{t \in [0, T]}$ defined by

$$M(t) := e^{\frac{t}{2}} Y(t)$$

Show that $(M(t))_{t \in [0, T]}$ is an $\mathbb{F}$ -martingale.

- iii. Derive a differential equation for the function $f(t) := \mathbb{E}[Y(t)]$, $t \in [0, T]$ and use it to find $f(t)$ explicitly.
- (d) [8 marks] Let $(W(t))_{t \in [0, T]}$ be an $\mathbb{F}$ -Wiener, let $x, y \in \mathbb{R}$, and let $(X(t))_{t \in [0, T]}$ and $(Y(t))_{t \in [0, T]}$ be given by

$$dX(t) = (-2X(t) + 3) dt + dW(t), \quad X(0) = x,$$

and

$$dY(t) = (-2Y(t) + 3) dt + dW(t), \quad Y(0) = y,$$

respectively. Show that for all $t \in [0, T]$, with probability one, we have

$$|X(t) - Y(t)|^2 = e^{-4t} |x - y|^2.$$

3. Let $T = 1$ and let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$ satisfying the usual conditions. Let $(W(t))_{t \in [0, T]}$ be a one-dimensional $\mathbb{F}$ -Wiener process on $(\Omega, \mathcal{F}, \mathbb{P})$. Consider the Black-Scholes model where there exist exactly one riskless asset $(B(t))_{t \in [0, T]}$ and a risky asset $(S(t))_{t \in [0, T]}$. The dynamics of $(S(t))_{t \in [0, T]}$ is given by

$$dS(t) = 5S(t) dt + 2S(t) dW(t), \quad S(0) = 1, \quad t \in [0, T],$$

where the risk-free interest rate is given by $r = 1$.

- (a) [9 marks] Let $\mathbb{P}^*$ be the risk-neutral probability measure of the above model and recall that under $\mathbb{P}^*$ the process

$$W^*(t) := \frac{5-1}{2}t + W(t)$$

is an $\mathbb{F}$ -Wiener process. Write down the stochastic equation (driven by W^*) that $(S(t))_{t \in [0, T]}$ satisfies under the measure $\mathbb{P}^*$ and write down the unique solution of this equation.

- (b) [12 marks] Set $\tilde{S}(t) = e^{-t} S(t)$.

- i. Show that $(\tilde{S}(t))_{t \in [0, T]}$ satisfies the equation

$$d\tilde{S}(t) = 2\tilde{S}(t) dW^*(t), \quad \tilde{S}(0) = 1.$$

- ii. Find $\mathbb{E}^*[\tilde{S}(t)]$ for all $t \in [0, T]$, where $\mathbb{E}^*[\cdot]$ denotes the expectation with respect to the measure $\mathbb{P}^*$.

- (c) [9 marks] Consider a derivative written on the risky asset which pays at the terminal time $T = 1$, the amount $|S(1)|^3$ to the owner. Find the price of this derivative at time zero (give a real number and show your workings).

The following can be used for (b) and (c): if $\theta \in \mathbb{R}$ and $X \sim N(0, 1)$, then the expectation of $\exp(\theta X)$ is given by $\exp(\frac{\theta^2}{2})$.